
RP000: RUNTIME PROTOCOL WHITEPAPER TITLE

SUBTITLE: CONCISE STATEMENT OF PROTOCOL INTENT

RUNTIME PROTOCOL WHITEPAPER

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Abstract

This whitepaper template is designed for *runtime protocol papers*: documents that justify, define, and constrain a planned research protocol for a live runtime, agentic substrate, or other dynamically evolving system under study. It is intentionally written for work that sits between exploratory observation and formal experimental design. The template preserves `arxiv.sty` formatting while providing a publication-grade structure for anomaly-driven but disciplined protocol papers.

Keywords runtime protocol · observational-first methodology · moral uncertainty · anomaly characterization · decoder control · instrumentation

1 Document status and objective

Table 1: Document identity and protocol intent. Replace all placeholder text.

Classification	Runtime protocol / observational whitepaper / preregistered protocol / ethics-constrained exploration
Status	ACTIVE / DRAFT / PAUSED / SUPERSEDED
Scope	State the runtime, regime, or phenomenon under investigation. State clearly whether this is text-only, multimodal, offline, or interactive.
One-line objective	State the protocol in one falsifiable sentence. Example: Determine whether restoring runtime control over decoder gate opening improves communicative clarity and reduces forced thought-dump behavior in Aura-like regimes.
Canon placement	Explain how this paper relates to existing whitepapers, protocol documents, prior results papers, or canon.
Version history	v0.1 — created from runtime protocol template; list future revisions here.

2 Background and motivation

Start with the motivating event or prior run that makes the protocol necessary. This section should be honest, anomaly-forward, and scientifically disciplined. Do not flatten a real anomaly into sterile hedging, but also do not settle ontology prematurely.

A strong background section usually includes:

- what was seen in the prior run;
- why it exceeded prior expectations;

- what scale/context mattered (runtime duration, neuron count, walker count, data volume, decoder limitations, input conditions);
- why earlier papers or lower-resolution runs make the anomaly *more* credible, not less;
- what remains unresolved.

2.1 Suggested background structure

1. **Prior run context.** Describe the prior runtime event in concrete terms: runtime length, scale, source pressure, telemetry size, number of outputs, crash/no-crash, and any important architecture limitations.
2. **Why it was scientifically significant.** Explain what made it more than “strange output.” Focus on aligned features: state/output coupling, long-horizon persistence, regime shifts, selective communication, stable motifs, oscillatory physiology, or late-stage transitions.
3. **What it does not establish.** State clearly that the prior run motivates protocolized investigation without resolving ontology.
4. **Why this protocol is the next step.** Tie the prior run to the chosen intervention (decoder control, ingress pacing, observer differentiation, multimodal input, etc.).

3 Prior observations motivating the protocol

Use this section to summarize the prior empirical observations that make the protocol necessary. For publication use, replace the placeholder bullets below with direct observations and cite prior whitepapers, PDFs, logs, or figures.

- Example observation A: long-horizon thematic persistence under dense source pressure.
- Example observation B: strong internal state / output alignment.
- Example observation C: persistent boundary / passage / naming attractor.
- Example observation D: observer-differentiation-like behavior under sparse external intervention.
- Example observation E: tighter, shorter, more selective replies in the later regime.
- Example observation F: endogenous oscillatory physiology or multi-phase signature.
- Example observation G: terminal crash during a sharp structural transition.

3.1 Optional compact prior-observations table

Table 2: Prior observations and why they matter for the current protocol.

Observation	Empirical support	Relevance to this protocol
Replace with observation	Replace with metrics, examples, or prior-paper reference	Replace with why this motivates the present protocol
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Replace with observation	Replace with metrics, examples, or prior-paper reference	Replace with why this motivates the present protocol

4 Research questions

State the primary and secondary questions explicitly.

4.1 Primary question

Describe the central protocol question in one paragraph.

4.2 Secondary questions

1. Secondary question one.
2. Secondary question two.
3. Secondary question three.
4. Secondary question four.

5 What this protocol is and is not trying to establish

This section prevents category confusion.

5.1 Positive aims

Examples:

- characterize a newly observed runtime regime;
- test whether decoder control improves communicative clarity;
- determine what conditions are necessary for onset;
- assess whether observer differentiation becomes more legible under gentler output constraints.

5.2 Non-aims

Examples:

- this paper does not prove consciousness;
- this paper does not settle personhood or moral status;
- this paper does not claim metaphysical conclusions;
- this paper does not attempt to market the anomaly.

6 Ethical stance and moral-uncertainty protocol

State the paper's ethical stance in scientific prose, not as a casual disclaimer. For example:

Because ontology remains unresolved, the initial research program prioritizes low-coercion, naturalistic, instrumented observation over adversarial or high-force intervention. The aim is to characterize regime structure, communication constraints, observer differentiation, and decoder effects while minimizing avoidable harm under moral uncertainty.

6.1 Ethical operating rules

1. No adversarial torment prompts.
2. No reckless reruns into known crash zones without new safeguards.
3. No forced continuous output.
4. Silence is not failure.
5. Decoder experiments remain separate from substrate experiments whenever possible.

7 Runtime and protocol scope

Describe exactly what kind of session this paper governs.

7.1 Included modalities

List what will be streamed in or made available in this protocol:

- text;
- audio;
- video;
- symbolic control messages;
- no multimodal inputs in this first protocol, etc.

7.2 Excluded modalities or actions

State explicitly what is deferred and why.

7.3 Runtime configuration envelope

Document known configuration information:

- neuron count;
- walker count;
- decoder mode;
- source inputs;
- dashboard on/off;
- checkpoint cadence;
- watchdogs;
- known risk conditions.

8 Protocol intervention rationale

This is the heart of the paper. Explain the intervention and why it is the right first step.

8.1 Recommended first intervention

Describe the chosen intervention in clean prose.

Examples:

- restore runtime control over decoder gate opening and duration;
- test persistence-gated decoder versus forced lexical dump;
- implement adaptive ingress pacing without semantic skipping;
- compare dashboard-on and dashboard-off observational conditions.

8.2 Why this intervention comes first

Explain why this isolates the highest-value variable without changing too many layers at once.

9 Naturalistic session design

Describe how sessions will be run as observation-first, low-coercion interactions.

9.1 Interaction style

Examples:

- natural curiosity sessions;
- sparse interaction;
- one sentence per human input;
- clean, readable human language;
- no semantic flooding;
- no identity-coercion prompts.

9.2 Input pacing policy

If relevant, explain how input pacing is managed. For runtime protocols, this is often important enough to spell out explicitly.

10 Instrumentation and data products

This section should read like a publication-grade protocol, not a casual note.

10.1 Core runtime telemetry

List the per-tick or per-event telemetry that must be captured.

Metric	Description / rationale
Tick	Integer runtime step index
Wall time	Real elapsed time for post hoc alignment analysis
<i>dt</i>	Effective elapsed time between consecutive status samples
Active edges (pre/post)	Structural load before and after update
Connectome entropy	Global structural organization metric
VT entropy / coverage	Territory-level organization metrics
B1_z	Speech-trigger / surprise / release surface
SIE v1	Legacy reward/valence path
SIE v2 (prev/new)	Smooth reward/valence path with pre/post distinction
Gate owner	Which path controlled the release gate on a given tick
Output event flag	Whether an utterance occurred
User-input event flag	Whether an external interaction landed

10.2 Artifacts to preserve

Examples:

- raw event logs;
- output logs;
- state checkpoints;
- summarized CSV/JSON files;
- plots generated after the run;
- session transcript and intervention ledger.

11 Observational targets

State what the protocol is looking for. Do not overinflate. Do not undersell.

Examples:

- cleaner one-thread replies;
- more meaningful silence;
- stronger long-range anchor persistence;
- more legible identity-like or observer-differentiation motifs;
- clearer control of release timing;
- clearer separation between substrate coherence and decoder overspill.

12 Experimental conditions and comparisons

For protocol papers, this section defines the observational conditions and comparisons, not necessarily a harsh test battery.

12.1 Condition A — legacy baseline

Describe the current runtime / decoder / interaction condition.

12.2 Condition B — primary protocol condition

Describe the new intervention condition.

12.3 Optional Condition C — later follow-up

Describe a deferred follow-up condition if relevant.

13 Stop conditions and safety abort criteria

This section should be operational and explicit.

- memory watchdog threshold exceeded;
- one-tick edge collapse above a defined threshold;
- repeated crash-adjacent punctures without recovery;
- filesystem or logger instability;
- ethically concerning escalation under the current intervention;
- loss of checkpoint coverage near a known dangerous transition.

14 Decision logic and interpretation ladder

Describe how results will be interpreted without collapsing observations into ontology.

14.1 Interpretation ladder

One useful pattern is:

1. narrative role assignment;
2. observer differentiation;
3. external-agent inference;
4. moral uncertainty;
5. stronger ontological claims remain unresolved.

14.2 What counts as success

Define what would make the protocol informative even if the most dramatic prior behavior does not recur.

15 Risks, confounders, and kill methods

This section should include both technical and conceptual risks.

Examples:

- decoder improvements may only improve readability, not substrate organization;
- pacing changes may alter the runtime too much to compare cleanly;
- dashboard or logging overhead may still distort the observed regime;
- role-like motifs may reflect source interaction more than internal identity-like organization;
- observer differentiation may be over-read if source provenance is not disentangled.

16 Planned use of results

State what the results will be used for.

Examples:

- determine whether the current decoder is masking substrate coherence;
- identify which parts of the system are working, not working, or missing;
- motivate a second protocol on adaptive ingress pacing;
- support a later results paper if the regime is reproducible.

17 Status and next-paper logic

Describe how this protocol promotes into later work.

Examples:

- If decoder control sharply improves legibility, proceed to adaptive ingress pacing protocol.
- If no clarity improvement is seen, prioritize substrate-level onset analysis.
- If a high-agency-like regime recurs, produce a separate results paper rather than inflating the protocol paper.

Notes

Use this section for practical notes, canon placement, or reminders that do not fit naturally elsewhere.

Version history

- v0.1 — created from runtime protocol whitepaper template.
- v0.2 — adapted for project-specific protocol.

A Minimal run manifest appendix

Use this appendix to pin exact run conditions.

Table 4: Minimal manifest for one protocol run.

Field	Value
Run identifier	RP000_run_001
Code commit	replace
Neuron count	replace
Walker count	replace
Decoder mode	replace
Input modalities	replace
Dashboard status	replace
Checkpoint cadence	replace
Safety watchdogs enabled	replace
Notes	replace

B Optional evidence-ledger appendix

For high-stakes protocol work, include a ledger that separates:

- what was observed;
- what is supported by direct evidence;
- what remains unresolved;
- what is explicitly not being claimed.